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Hiring Manager
Oklo Inc

Dear Hiring Manager,

I am writing to apply for the Mechatronics Engineer position at Oklo. As a UC Berkeley M.Eng. graduate in Robotics with hands-on experience programming industrial robotic arms for fuel rod testing, I bring direct overlap between my background and Oklo's need for automated robotic material-handling systems in its Fuel Recycling Facility.

That fuel-handling experience shaped my approach to automation in constrained environments. At Xiaoxian AI, I developed embedded C control programs for a six-axis robotic arm performing automated multi-axis motion on nuclear fuel rods, then integrated IR sensors, microswitches, and pressure transducers into the workflow—building the kind of sensor-driven safety monitoring that unmanned operations demand. Converting spent fuel into clean energy through pyroprocessing requires exactly this: electromechanical systems that perform reliably when human intervention is not an option.

That principle carried into my subsequent prototyping and iteration work. On a remote-mining excavator R&D project, I designed an ESP32-based Wi-Fi remote control system achieving over 95% task repeatability, then consolidated five circuit boards into a single 4-layer PCB to harden the electronics for field use. At Berkeley, I engineered a magnetically coupled drive for a vacuum dryer that eliminated direct mechanical interfaces in a sealed chamber—a design philosophy directly transferable to leak-tight actuation in radiological environments. Across both projects, I delivered working hardware through rapid iterative cycles, completing three prototype revisions in two months on the excavator alone.

I recognize I lack direct high-radiation environment experience. However, my combined background in robotic fuel-handling, remote-operated systems, and sealed-environment engineering positions me to contribute quickly to Oklo's first-of-kind facility. I would welcome the opportunity to discuss how my skills align with your team's mission.

Sincerely,
Felix Hu